Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Engineer with 3 years of industry experience building and maintaining ETL/ELT pipelines, curated datasets and reporting in a regulated financial-services setting: **Python** and **SQL** ingestion into **PostgreSQL**, scheduled **AWS** pipelines (**S3, Lambda, EventBridge, SageMaker**) and **Power BI** dashboards used by analysts. Translated requirements from investment, compliance and client-service stakeholders into reliable data products, and acted as de facto team lead for up to 4 interns. Recently completed an MSc in Data Science at the University of Bristol, including a data-preprocessing pipeline with integrity checks for a dissertation project in partnership with AstraZeneca. Seeking a Data Engineer role building maintainable data products on Azure and Databricks.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Pipelines & ETL/ELT

- Wrote **Python** ingestion scripts against the enterprise-licensed **Morningstar API**, covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity), and **integrated** it with the legacy data previously held in analysts' **Excel** files.
- Built a scheduled **Python** pipeline on **AWS SageMaker** (**EventBridge** and **Lambda**, 9 AM daily) that parses over **300 daily emails** from credit rating agencies, fuzzy-matches them to the companies analysts track, emails a daily consolidated summary and stores an **Excel** output in **S3**, **saving 60 manual analyst hours a month**.
- Troubleshot the pipeline when one rating agency began sending emails as images and the **AWS Textract** parser broke; replaced it with **Azure OpenAI**, which also improved accuracy.

Data Modelling, SQL & Curated Datasets

- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) for the internal analytics platform, to the analysts' requirements and guidance.
- Developed **Optimized SQL** queries for data preprocessing, and **SQL Functions** and **Procedures** for efficient data insertion.
- Served the same datasets to a GenAI application whose **Python** functions run SQL queries for fund and fund-house comparisons.

BI & Self-Service Analytics

- Developed **Power BI** dashboards for fortnightly credit report analysis for the investment team, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Recreated the **Power BI** dashboards in the **ReactJS** frontend because licensing was too costly to scale to the nationwide client service team, giving the wider organization self-service access to the data at lower cost.
- Analyzed customer data using **Python Pandas**, which was used across data jobs, backend APIs and RAG pipelines, and identified buying patterns using **Market Basket Analysis** (**Apriori Algorithm**) for the marketing team.

Deployment, Governance & Stakeholder Delivery

- Designed the infrastructure architecture for internal analytics platforms and applications on **AWS** using **EC2, S3, ECR and Lambda**, with **Docker** microservices and a one-click **Jenkins** and **GitLab** CI/CD pipeline (**Git** repositories with dev and prod branches).
- Restricted access to company private IPs using **AWS Security Groups**, implemented **RBAC** with **LDAP** authentication, and worked with the security team to take the applications from repeatedly failing to passing **VAPT**.
- Worked directly with Client Service, Compliance and Investment stakeholders to turn requirements into working tools, including **RAG** applications (**LlamaIndex**, **Azure OpenAI**, **pgvector**) that Compliance reviewed; acted as de facto team lead for up to 4 interns; recognised internally with **Best Team Project** twice (2024, 2025).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I worked on the data preprocessing, scoring algorithm and RAG layer. A pipeline turns 17 raw inputs into 14 clean, joined tables (CSV and Parquet) with identity resolution, duplicate and missingness handling, integrity checks and an audit manifest; ranks cancer cell lines by target inclusion and exclusion genes; validated with feedback from AstraZeneca scientists.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. (Made for the Swindon Borough Council Fuel Poverty Hackathon.) | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker), Power BI, Azure OpenAI, Docker, Jenkins, GitLab, Git, Nginx, Linux, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (schema design, SQL functions and procedures, pgvector), SQLite